



St. MARTIN'S ENGINEERING COLLEGE

(UGC Autonomous)

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ASSIGNMENT ANSWER BOOKLET

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Course : B.Tech

Branch : IT

Year : I/II/III/IV

Sem : I / II

Assignment : I / II

Subject Name : Information Security

Last Date of Submission: 17/11/25

MARKS AWARDED

Q.No.	a	b	Total
1			
2			
3			
4			
5			
Total			

Actual Date of Submission : 12/11/25

V. Manish

Signature of the Student

Signature of the Evaluator

Start Writing from Here

1) Explain various authentication protocols in Information Technology.

Ans An authentication protocol is a type of computer communications protocol specially designed for transfer of authenticated data b/w two entities. It verify the identity of users, systems (or) Entities before getting access to data.

PAP Password:-

1) Kerberos:-

It is a protocol that aids in network authentication. This is used for validation client/servers during a network. Employing a cryptographic



Key. It is designed for executing strong authentication while reporting to applications. The overall implementation of Kerberos protocol is openly available by MIT and used mainly in produced products.

2) Light weight Directory Access protocol (LDAP):-

LDAP refers to Lightweight Directory Access protocol. It is a protocol that is used for determining any individuals, organizations, and other devices during a network regardless of being public (or) corporate internet. It is practised as directories as a service & it is the grounds for Microsoft buildings.

3) OAUTH:-

OAUTH as the name suggests it is an authorization framework that promotes granting limited access to user on its account through an HTTP service. When a user requests access to resource on API call is made & after the authentication taken is passed.

4) SAML:-

SAML stands for Security Assertion Markup Language which is based on XML based authentication data format which provides the authentication b/w an identity provider. The authentication data format which provides the ~~authen~~ authorization b/w the identity provider. It serves as a product of the OASIS security services.



RADIUS:-

RADIUS stands for Remote Authentication Dial-in User Service

It is a network protocol that provides sufficient centralized Authentication Accounting & Authorizing for the users that can use network services.

2) Briefly Explain the various services provided by PGP.

Ans) PGP is a data Encryption & decryption program that provides cryptographic privacy and authentication for Email & file communication. It combines symmetric key Encryption & public key Encryption for secure communication.

i) Authentication

→ Ensures that the message usually comes from the claimed sender.

→ PGP uses digital signatures for authentication.

ii) Confidentiality:- Ensures that the message content remains secret & can be read only by the internet recipient.

PGP achieves this by encryption the message using a session key (symmetric Encryption like AES)



3) Compression:-

Before encryption, PHP compresses the message by using zip like algorithms.

4) Segmentation:-

When a message is too large, PHP divides it into smaller segments for transmission.

Each segment is reassembled at the receiver end.

5) Email Compatibility:-

PHP converts binary data (Encrypted content) into ASCII using radix 64 encoding.

This makes encrypted message compatible with email systems that handle only text.

3) Discuss about IP security (IPsec) Architecture

Ans IP security Architecture:-

IP security is a framework of open standards developed by the IETF to ensure secure communication over IP network (like the internet).



It operates at the network layer (layer 3) and provides end to end security for IP packets.

Components of IPsec Architecture:-

Security Associations:-

1) A security association defines the parameters & keys used for secure communication b/w two devices.

It is one way (unidirectional) relationship.

Security policy database:-

2) The SPD contains rules that determine how to handle outgoing or incoming traffic.

Discard: Drop the packet

Bypass: Send the packet without IPsec.

Security Association Database

3) The SAD stores details of active SAs including encryption & authentication algorithms, key sequence numbers.

4)



Differentiate short notes about SSL & its protocol.

Ans)

SSL (Secure Socket Layer) is a cryptographic protocol designed to secure communication on a computer network such as the Internet.

Here are the short notes differentiating SSL itself and main protocols with SSL.

Overview:

SSL is a standard security protocol that establishes

Encryption (link) between a web server & a browser ensuring all data transferred remains private & secure.

It uses digital certificates to verify the identities of communication parties & enable secure transmission of sensitive information like passwords & credit card numbers.

HTTPS is HTTP over SSL (or its successor) the websites

using SSL show "https" in the URL's.

Main SSL protocols:-

SSL encompasses several sub protocols, each handling a specific security aspect.

Record protocol:- Ensures the confidentiality and integrity

of data by fragmenting, compressing & applying a message



Authentication code (MAC)

Handshake protocols:-

Establish the SSL session authenticate parties & negotiates Encryption algorithm & session key through a series steps.

5) Explain the Intrusion detection system techniques & various types of ~~to~~ firewalls.

→ An intrusion detection system (IDS) is a security mechanism that monitors network or system activities for malicious actions, policy violations or anomalies.

It alerts administrators when suspicious behaviour detected.

Techniques of intrusion detection:-

It is mainly divided into these categories

Signature Based Detection:-

Concept:- Detects attacks by comparing network traffic

(a) System activities with predefined patterns or signature 7



of known threats

2) Anomaly Based Detection:-

Concept:- build a base line of network / system activity

Any deviation from this base is flagged as suspicious activity.

3) Hybrid detection:-

Concept:- combine signature based and anomaly based methods - to leverage the benefits of both.